IEEE ClimateChain Global Hackathon · Track 3: Sustainable Supply Chains

PickPay

Every kilo of plastic paid once, claimed once.

Waste pickers in Lagos are paid within seconds for plastic that AI grades from a photo. Recycling credits exist only after a recycler confirms the plastic, and each kilo can be claimed by exactly one brand, enforced by a Solana program.

Code	https://github.com/Don-Vicks/pickpay
Demo video	https://youtu.be/wqHe-604t4c
Program (devnet)	47EHi4xhP3o6T7TUsMVRNMse4Xdcs1zRHLXxSpncgLTz
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Contents: 1. Problem, climate impact and roadmap · 2. Architecture and technical spec · 3. Security and data-integrity assessment

Problem, climate impact and roadmap

Problem statement

Nigeria generates an estimated 2.5 million tonnes of plastic waste a year, about 870,000 tonnes of it in Lagos (ThisDay, 2024). A 2021 survey by UN-Habitat and the Lagos Waste Management Authority found that Lagos collects 48% of its municipal solid waste, handles 8% in controlled facilities and recovers 8% for recycling; the rest is left uncollected or dumped (UN-Habitat Waste Wise Cities factsheet, Lagos). These figures are estimates; Nigerian waste data is thin, and the sources say so.

Two things keep plastic from reaching recyclers:

1. **Collection doesn't pay reliably.** Informal pickers gather much of what is recovered, sell by weight to middlemen, and are paid in cash, late, on scales nobody records. Recyclers report they can't get enough feedstock even when they pay anyone who brings it (Africa Live).
2. **Producers now owe proof, and the proof is weak.** The National Environmental (Plastic Waste Control) Regulations 2026 make producers and importers responsible for recovering the plastic they sell, and require PET packaging to contain 25% recycled PET by January 2028 and 50% after that (ThisDay, Sept 2026). A brand's evidence today is a report from the collector it paid. Nothing stops the same tonne being reported to two brands, or reported without ever reaching a recycler.

Solution

PickPay connects the two. A hub weighs and photographs each bag; AI grades it; the picker is paid in the same Solana transaction that records the bag. When the hub ships a bale, only the receiving recycler's signature creates recovery credits, capped at the bale's weight. Brands retire those credits against their EPR obligations, and the program rejects any claim on kilos already retired. Anyone can trace a claim back to the bags it came from.

Users and who pays

User	Gets	Pays
Waste pickers	Payment in seconds at a fair, consistent grade-based price; a record of their work	Nothing
Collection hubs (buy-back centres)	Faster throughput, no cash handling, a public track record	Nothing in the pilot; a small fee per tonne later

User	Gets	Pays
Recyclers	More and cleaner feedstock, graded before it arrives	Nothing
Brands and producer-responsibility organisations	EPR evidence regulators and auditors can check without trusting a vendor	A fee per verified tonne claimed

Climate impact assessment

How PickPay reduces emissions: every kilo of PET that reaches a recycler instead of a dump or a fire (a) displaces virgin PET resin and (b) avoids the emissions of open burning and dumping. Only (a) is counted below, because it is the better-measured of the two.

Evidence: NAPCOR's life-cycle analysis (Franklin Associates, 2020) finds that replacing virgin PET with recycled PET cuts greenhouse-gas emissions by about 60% (Recycling Today). A per-kilo estimate puts virgin PET at about 6.5 kg CO2e/kg and recycled PET at 4.3 kg CO2e/kg (Carbonfact), a saving of about **2.2 kg CO2e per kg** recycled. That source's boundaries are unverified and its 34% saving is more conservative than NAPCOR's 60%, so it is used here as the conservative figure.

Projection (illustrative, conservative assumptions):

Scenario	PET recovered per year	CO2e avoided per year	Paid to pickers per year
One pilot hub, 1 t/day, 300 days	300 t	≈ 660 t	≈ ₦90m at ₦300/kg
10 hubs in Lagos	3,000 t	≈ 6,600 t	≈ ₦900m
100 hubs across Nigeria	30,000 t	≈ 66,000 t	≈ ₦9bn

Assumptions: PET only (HDPE and mixed plastic not counted), the conservative 2.2 kg CO2e/kg saving, avoided burning and dumping not counted, prices at the demo rate of ₦300/kg for clean PET. Ten hubs would recover about 0.3% of Lagos's plastic waste, so the impact depends on growing the network, not on one site. The ledger itself is the measurement: every figure in this table becomes an on-chain number once hubs are live.

User testing: none yet. The first pilot (below) is where it happens; the metrics to track are payment time, grade agreement with recycler intake, the collected-to-accepted gap per hub, and picker retention.

Adoption strategy

1. **Start where the plastic already moves.** Lagos already has buy-back centres and aggregators. PickPay replaces their cash box and notebook with a phone, a scale and a payout float; nothing about the physical flow changes.
2. **Recyclers first among the institutions.** Recyclers want steadier, graded feedstock and lose nothing by signing intake; their signature is what makes the credits valuable.
3. **Brands follow the evidence.** Under the 2026 regulations, producers and their producer-responsibility organisations need recovery evidence they can defend. A per-tonne fee for verified, single-claim credits funds hub floats and operations.
4. **Pickers without bank accounts or smartphones** are paid to a balance tied to their phone number; a withdrawal flow, a naira off-ramp and USSD come in the pilot.

Scalability and sustainability roadmap

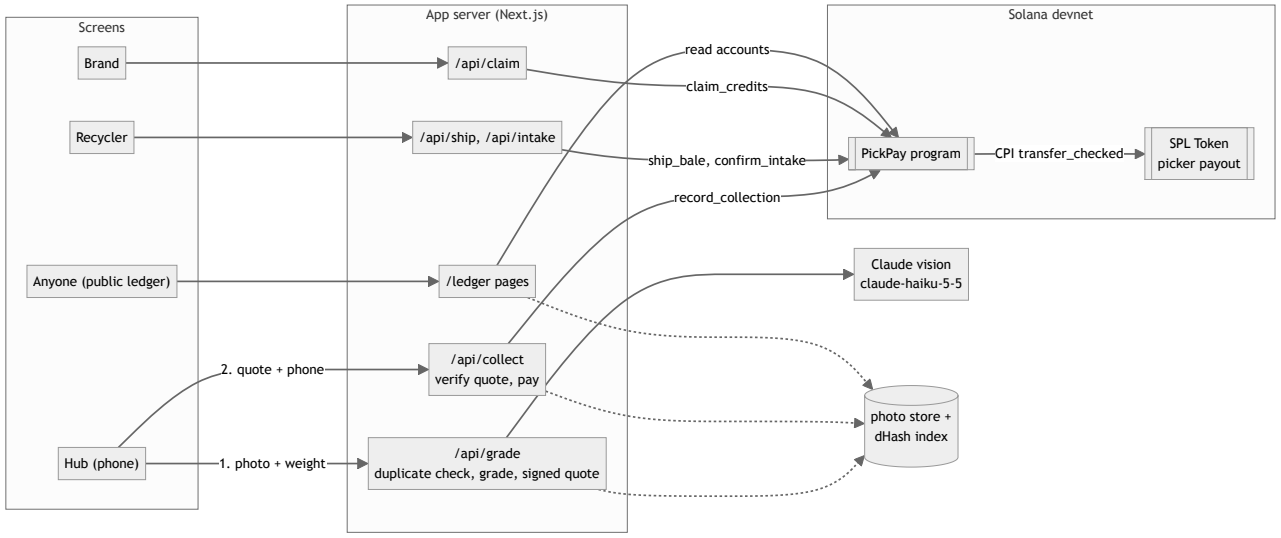
Phase	When	What
Pilot	Months 1–3	One Lagos hub and one PET recycler; real USDC payouts with a naira off-ramp; picker withdrawals and USSD; grade accuracy measured against recycler intake
Brands	Months 3–6	First producer-responsibility organisation or brand claiming credits; monthly public reports generated from the ledger; per-tonne fee live
Network	Months 6–12	10 hubs and 2–3 recyclers; staked hubs with slashing on disputed bales; scales that sign weights; an indexer for chain reads
Beyond	Year 2	Other waste streams (HDPE, e-waste, batteries) on the same attestation pattern; other West African cities

Technical scaling: hubs are independent accounts, so they write in parallel; each bag costs about 0.0033 SOL of reclaimable rent and one transaction fee. **Financial sustainability:** brand fees per verified tonne cover hub floats and operating costs; pickers and recyclers pay nothing.

PickPay architecture and technical spec

PickPay has three parts: an Anchor program on Solana that holds every fact and enforces the rules, a Next.js app that the hub, recycler and brands use, and Claude vision, which grades bag photos. There is no database. The app reads chain state for every screen; the hub keeps only bag photos and their perceptual hashes.

System



On-chain program

Program ID (devnet): `47EHi4xhP3o6T7TUsMVRNMse4Xdcs1zRHLXxSpncgLTz`. Anchor 1.2, built for SBPF v1 (`anchor build --arch v1`). Weights are stored in grams and payouts in the payout mint's base units (6 decimals).

Accounts

Account	Seeds	Holds
`Hub`	`"hub", authority`	authority, payout mint, escrow, `collected_g`, `shipped_g`, `paid_total`, `owed_total`, `collection_count`, `bale_count`, `next_unbaled`
Escrow (SPL token account)	`"escrow", hub`	the hub's payout float; owned by the hub PDA, so only the program can move it

Account	Seeds	Holds
`Collection`	`"collection", hub, index (u64 LE)`	one bag: picker hash, paid-to wallet (or default for accrued), weight, polymer, contamination (bps), amount, photo SHA-256, time
`PhotoSeal`	`"photo", photo_sha256`	exists once per photo hash, so the same photo can't back two bags
`Picker`	`"picker", hub, picker_hash`	per-picker totals and the balance owed to pickers without a wallet
`Bale`	`"bale", hub, index`	weight shipped, the recycler named to receive it, the range of bags it carries, intake flag
`CreditBatch`	`"credits", bale`	recovery credits from a recycler's intake: accepted g, claimed g, claim count
`Claim`	`"claim", credit_batch, index`	one brand retiring part of a batch

`picker_hash` is SHA-256 of a server-side salt and the E.164 phone number. Phone numbers never reach the chain.

Instructions

Instruction	Signer	Effect	Rejects
`init_hub`	hub authority	creates `Hub` and its escrow token account	
`record_collection(args)`	hub authority	creates `Collection` and `PhotoSeal`, updates `Picker` and hub totals; transfers `amount` from escrow to the picker's token account in the same instruction, or adds it to `Picker.owed` if no token account is passed	zero weight; contamination > 10,000 bps; payout account on the wrong mint; a photo hash already sealed
`ship_bale(weight_g, recycler)`	hub authority	creates `Bale` for the bags recorded since the last bale; `shipped_g += weight_g`	`ExceedsCollected` if `shipped_g` would pass `collected_g`

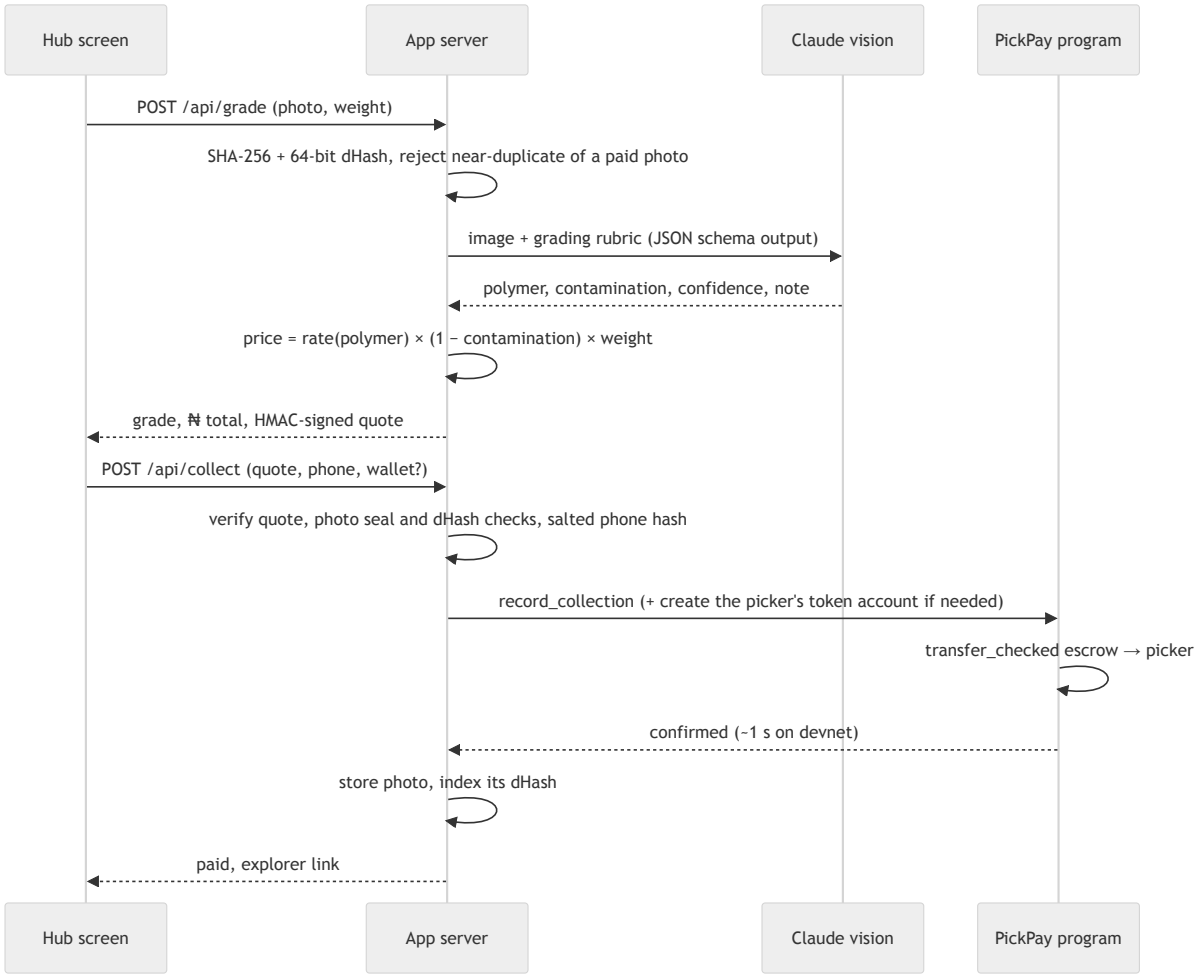
Instruction	Signer	Effect	Rejects
`confirm_intake(accepted_g)`	the bale's recycler	creates the bale's single `CreditBatch` with `accepted_g`	`WrongRecycler`; `ExceedsIntake` if `accepted_g > bale.weight_g`; a second intake (batch already exists)
`claim_credits(weight_g)`	brand	creates `Claim`; `claimed_g += weight_g`	`AlreadyClaimed` if `claimed_g` would pass `accepted_g`

The invariant **collected ≥ shipped ≥ credited ≥ claimed** follows: shipping is capped by collection per hub, credits by the bale they came from, and claims by the batch. All arithmetic is checked (`Overflow` on failure).

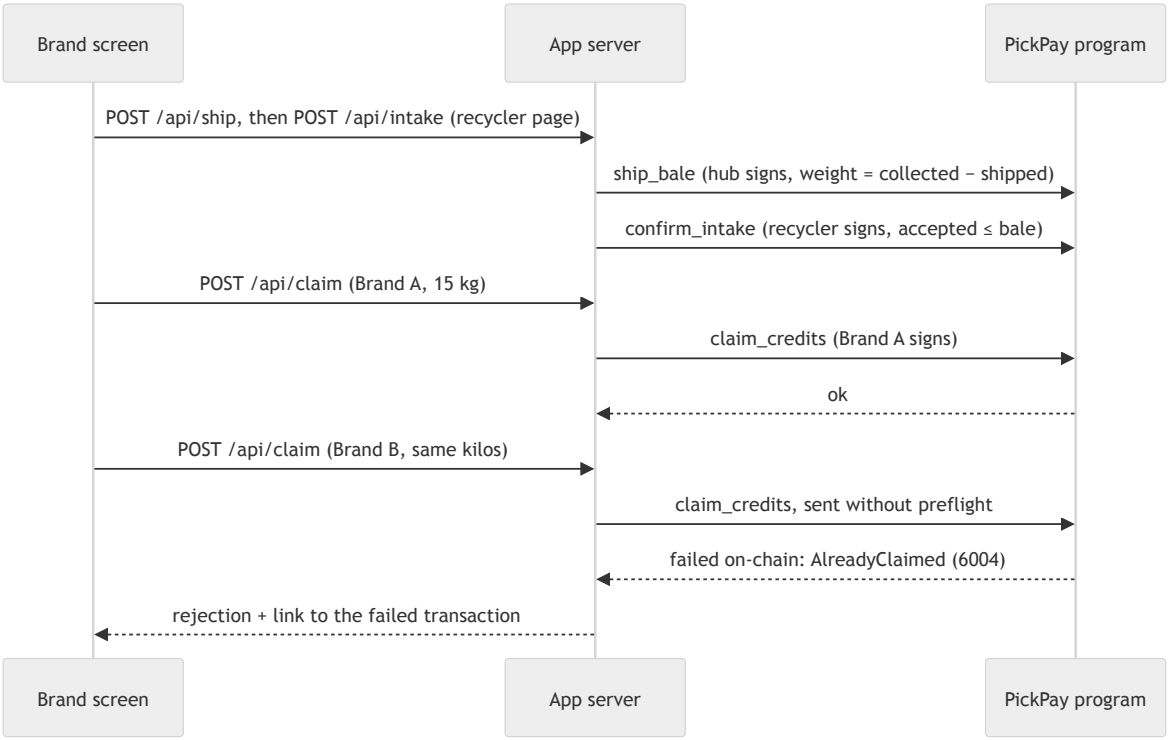
Tests

`programs/pickpay/tests/` runs 11 LiteSVM tests against the compiled program: escrow setup, payout to a wallet in the same transaction, accrual without a wallet, duplicate photo, invalid inputs, wrong payout mint, the full bag → bale → credit → claim chain, `ExceedsCollected` (including across two bales), `ExceedsIntake` and double intake, wrong recycler, and a second brand claiming retired credits.

Collecting a bag



Recycling and claiming



Claims skip preflight on purpose: the rejection lands on-chain, so the explorer shows the program's own error rather than a server-side check.

App

Route	Role
`/`	How it works in three steps, with live totals
`/hub`	Step 1, a mobile screen for the hub agent: phone, optional wallet, weight, photo → grade → pay; recent bags
`/recycler`	Step 2: the hub sends unshipped bags as a bale; the recycler enters the weight that arrived and confirms
`/brand`	Step 3: plastic still available to claim, claims as Brand A or B, plain-language results
`/ledger`, `/ledger/[claim]`	Public trace of every claim down to the bags, from chain state
`/api/grade`	Duplicate check, Claude grading, pricing, signed quote
`/api/collect`	Verifies the quote, sends `record_collection`, stores the photo
`/api/ship`, `/api/intake`, `/api/claim`	Hub, recycler and brand transactions (intake and claims skip preflight, so the program's refusal is on-chain)

Route	Role
`/api/ledger`, `/api/bags`, `/api/photo/[sha]`	Chain reads for the demo hub, recent bags, and paid photos

- **Grading:** `claude-haiku-5-5` (the cheapest current Claude model, about 3 s per photo) at low effort with a JSON schema output; a refusal returns a clear error to the hub. The model sets polymer and contamination; pricing is deterministic integer maths in `app/src/lib/pricing.ts` (demo rates: PET ₦300/kg, HDPE ₦250/kg, mixed ₦100/kg, discounted by contamination).
- **Signed quotes:** `/api/grade` signs `{photo SHA-256, dHash, weight, polymer, contamination, amount, expiry}` with an HMAC key derived from the hub's secret; `/api/collect` pays only a valid, unexpired quote.
- **Duplicate photos:** exact reuse is stopped on-chain by `PhotoSeal`; near-duplicates (≤10 of 64 dHash bits differ) are stopped by the hub's index before grading and again before payment.
- **Client:** `@solana/kit` 8.5 with a Codama-generated client (`pnpm generate:client`); PDAs whose seeds use instruction data or counters are derived in `app/src/lib/pdas.ts`.
- **Keys:** the demo server holds the hub, recycler and brand keys (`.keys/`, never committed). In production each party signs on its own device; see security.md.

Scaling notes

- Each bag is one transaction that creates about 210 bytes of account data (Collection 171 B, PhotoSeal 40 B, plus a 73 B Picker on a picker's first visit), roughly 0.0033 SOL of rent at current rates; rent is the main cost and can be reclaimed by closing settled accounts.
- Hubs are independent PDAs, so many hubs write in parallel without contending for one account.
- Chain reads use `getProgramAccounts` filtered by discriminator; beyond a few thousand accounts an indexer (Geyser or a webhook service) should replace it.

Security and data-integrity assessment

PickPay's claim is narrow and checkable: **a recovery credit can't exist without a recycler's signature, and a credited kilo can't be claimed twice.** This document sets out what the system guarantees, what it doesn't, and how each weakness is handled now or planned. The program has not been audited; it runs on devnet only.

What is guaranteed by the program

These hold whatever the app server, the hub or the AI does, because the program checks them on every transaction:

1. **No credit without intake.** Credits are created only by `confirm_intake`, which must be signed by the recycler named on the bale when it shipped (`WrongRecycler` otherwise), and only once per bale (the credit batch PDA is derived from the bale).
2. **No credit beyond what was shipped.** `accepted_g ≤ bale.weight_g` (`ExceedsIntake`), and a hub's total shipped weight can't exceed its recorded collections (`ExceedsCollected`).
3. **No double claim.** `claimed_g + weight_g ≤ accepted_g` for every claim (`AlreadyClaimed`). All brands draw from the same account, so two brands, or one brand twice, can't count the same kilos.
4. **One photo, one paid bag.** `PhotoSeal` is a PDA seeded by the photo's SHA-256; a second bag with the same bytes fails to create it.
5. **Payment and record can't drift apart.** The picker's payout is a `transfer_checked` CPI inside `record_collection`; the bag record and the payment succeed or fail together. The escrow is owned by the hub PDA, so only the program can move it.
6. **Arithmetic and accounts.** Every counter uses checked arithmetic (`Overflow`). Accounts are bound by seeds and `has_one` constraints (hub authority, mint, escrow); the payout account must use the hub's mint; every state change requires the right signer.

These are covered by 11 LiteSVM tests against the compiled program, and the double-claim rejection has been reproduced on devnet (failed transaction, error 6004).

Threat model

#	Threat	Who	Mitigation now	What remains
1	**Hub inflates weights** to pay friends or pad its numbers	hub	The hub pays inflated bags from its own escrow, so it funds its own fraud. Inflated weight never becomes credit: the recycler weighs the bale independently and can accept less than shipped. The gap between collected and accepted is public per hub.	A hub and recycler who collude can still invent kilos. Planned: hubs post a stake that a recycler or auditor can slash on a disputed bale, connected scales that sign weights, and random audits of high-gap hubs.
2	**Same photo paid twice**	hub or picker	Exact copy: rejected on-chain by `PhotoSeal`. Resized, recompressed or brightened copy: rejected by a 64-bit perceptual hash (≤10 differing bits) before grading and again before payment.	A cropped or rotated copy, or a fresh photo of the same bag, can pass. The recycler's intake cap (threat 1) bounds the damage; the index lives on the hub, not on-chain.
3	**Tampered payout** from the browser	anyone with the hub screen	The price is computed on the server. `/api/grade` signs the quote (photo hash, perceptual hash, weight, polymer, contamination, amount, 10-minute expiry) with an HMAC key derived from the hub's secret; `/api/collect` pays only a valid, unexpired quote. Tested: an edited amount is rejected.	Whoever holds the hub key can still sign anything (threat 7).
4	**Quote replay**	anyone	A quote is bound to one photo hash; after it's paid, `PhotoSeal` and the duplicate index reject a second payment.	None known.

#	Threat	Who	Mitigation now	What remains
5	**Misleading the AI grade** (e.g. text in the photo saying "clean PET")	picker	The model returns a strict JSON schema against a fixed rubric; the price is bounded by the clean rate for the polymer × weight, so a wrong grade can at most move a bag to the best clean rate. The hub agent sees the grade, confidence and note before paying.	Systematic over-grading. Planned: log grades against recycler intake to measure per-hub and per-model accuracy; a second-opinion grade above a value threshold.
6	**Recycler over-accepts** to create credits	recycler	Capped at the bale weight; only the named recycler can sign; one intake per bale.	A recycler accepting a bale it never received, in collusion with a hub (threat 1).
7	**Key theft or abuse**	attacker, insider	Demo only: the server holds the hub, recycler and brand keys in `.keys/` (gitignored) so one machine can run the demo.	A stolen hub key can pay out the escrow through `record_collection`. Production plan: each party signs on its own device or wallet; hub keys on a managed device with a low escrow float topped up daily; a per-bag payout cap and daily limit in the program.
8	**Program upgrade**	deployer	Devnet only; the upgrade authority is the developer's key.	Production: upgrade authority held by a multisig, then frozen once stable; verifiable builds.
9	**Server-side shortcuts**	app operator	Nothing that matters is enforced only by the server: claims are sent without preflight so the program, not the server, decides; every ledger figure is read from chain state.	The duplicate-photo index and quote signing are server-side by design (threats 2, 3).
10	**Fake ledger page**	anyone	Each figure links to the account on a public explorer; a bag photo is marked verified only if its SHA-256 matches the hash stored on-chain. The trace page accepts only accounts owned by the program with the `Claim` discriminator.	Readers must use the explorer links to trust a copy hosted elsewhere.

Privacy

- **Pickers:** only `SHA-256(salt : E.164 phone)` goes on-chain. With the salt, phone numbers are easy to brute-force (the number space is small), so the salt is a server secret (`PICKER_SALT`) per deployment and must not be published. Bags by the same picker are linkable to each other on-chain under that pseudonym.
- **Photos:** stored on the hub, not on-chain; only their hashes are public. Bag photos should not show people; the hub screen asks for a photo of the bag.
- **Brands and recyclers:** identified by their public keys, which is the point: claims are meant to be public.

Data integrity, end to end

Fact	Source of truth	How a reader checks it
A bag was weighed, graded and paid	`Collection` account + the payout transaction	Explorer link on the ledger page
The photo is the one graded	SHA-256 in `Collection.photo_hash`	Ledger shows "photo matches the blockchain record ✓" after hashing the stored file
A recycler received the plastic	`CreditBatch` created by the recycler's signature	The intake transaction's signer
A brand's claim is unique	`CreditBatch.claimed_g ≤ accepted_g`	Batch account; the failed transaction for any over-claim

Known gaps, in order of priority

1. Hub–recycler collusion (threats 1 and 6): staking and slashing, signed scales, audits.
2. Key custody (threat 7): per-party signing, payout caps, small floats.
3. Pickers' accrued balances have no withdrawal instruction yet; funds stay in the hub's escrow until one is added.
4. No independent audit of the program.